3.5 Higher-Order Derivatives — teacher key

Question: What does it mean to take the derivative of a derivative?

Self-paced lesson

No clock, no trimming, no skipping, no end mark, no stamp. Every tab is required in order; progress saves on the device so a student can stop and finish the next day.

Warm-up review (3 easy items)

- The derivative of x^3 is... **$3x^2$.** Power rule: bring the 3 down and lower the power by 1.
- The derivative of $\sin x$ is... **cos x.** The derivative of sine is cosine.
- If $f'(x) > 0$ on an interval, then f is... **increasing.** A positive slope means the graph of f rises.

Explore stations

Four hands-on stations (X1, X2, X3, X4). Station answers are in the teacher view of the lesson page (add `#key` to its address).

Explore: what did you see

- In Station 1, where the red tangent on f was flat, the graph of f'... **was at 0, on the x-axis.** A flat tangent has slope 0, and f' is the slope. So f' equals 0 there.
- In Station 1, at the lowest point of f', the value of f'' was... **0.** The tangent on f' is flat at its lowest point, and f'' is the slope of f'. So $f'' = 0$.

Mid-lesson check (Huddle tab)

- If f is a cubic polynomial, then f''' is... **a constant.** Each derivative lowers the degree by one: cubic, parabola, line, constant.
- The slope of the graph of f' at $x = 3$ equals... **f?(3).** f'' is the derivative of f', so it is the slope of the graph of f'.

Targets

- I can name f?, f? and f? and say what each one tells me.
- I can find a second or higher derivative with the usual rules.
- I can find d^2y/dx^2 at a point on an implicit curve.

Predictions

- The derivative of x^2 is $2x$. The derivative of $2x$ is... **2.** The line $2x$ has slope 2, so differentiating again gives 2.
- Keep differentiating $\sin x$: $\sin x$, $\cos x$, $-\sin x$, $-\cos x$, and then... **sin x.** The next derivative of $-\cos x$ is $\sin x$. The pattern repeats every 4 derivatives.

Huddle script (answers in green)

Definitions

- **Higher derivative:** Differentiate **again**. The second derivative is f?(x) or d^2y/dx^2. The third is f?(x).
- **What f? means:** f? is the **slope** of f?. It tells how the slope of f is **changing**.
- **Reading the sign:** If f? > 0, the slope of f is **increasing**. If f? < 0, it is **decreasing**.
- **Patterns:** A polynomial loses **one** degree each time. sin x repeats every **4** derivatives.

Worked example

Let f(x) = x? – $3x^2$ + 2x. Find f? and f?.

- First derivative: f?(x) = $4x^3$ – 6x + 2
- Second derivative: f?(x) = **12**x^2 – 6
- Third derivative: f?(x) = **24**x
- The fourth derivative is the constant **24**

Label the three graphs

1. The cubic is the graph of **f**.
2. The parabola is the graph of **f?**.
3. The straight line is the graph of **f?**.

Every level, with answers

Second derivative

Level 1. Find the second derivative.
Answer: $24x + 8$ $f'(x) = 12x^2 + 8x + 3$, so $f''(x) = 24x + 8$.

Level 2. Find the second derivative.
Answer: $4e^{2x}$ $f'(x) = 2e^{2x}$. Differentiating that again gives the answer.

Level 3. Find the second derivative.
Answer: $\left(2 + 4x^2\right) e^{x^2}$ $f'(x) = 2x\, e^{x^2}$. Differentiating that again gives the answer.

Level 4. Find the second derivative.
Answer: $\left(x^2 + 4x + 2\right) e^{x}$ $f'(x) = \left(x^2 + 2x\right) e^{x}$. Differentiating that again gives the answer.

Higher derivative at a point

Level 1. $f(x) = e^{2x}$. Find $f^{(4)}(0)$.
Answer: 16 Each derivative multiplies by 2. After 4 derivatives: $2^4 e^{2x}$. At $x = 0$ that is 16.

Level 2. $f(x) = \sin x$. Find $f^{(99)}(0)$.
Answer: -1 The pattern repeats every 4. 99 leaves remainder 3, so $f^{(99)}$ is the same as $f^{(3)}$. At $x = 0$ the value is -1.

Level 3. $f(x) = \cos(2x)$. Find $f^{(6)}(0)$.
Answer: -64 Each derivative multiplies by 2, so the size is $2^6 = 64$. The sign follows the cos pattern: $-$. The answer is -64.

Level 4. $f(x) = \cos(3x)$. Find $f^{(4)}(0)$.
Answer: 81 Each derivative multiplies by 3, so the size is $3^4 = 81$. The sign follows the cos pattern: $+$. The answer is 81.

Read the second derivative

Level 1. $f''(x) < 0$ on an interval means the slope of f is...
(A) increasing **(B) decreasing** (C) constant (D) undefined
Answer: B f'' is the slope of f'. Positive: f' goes up. Negative: f' goes down. Zero: f' stays level.

Level 2. $f'(x) = x^2 - 4x - 4$. At $x = 4$, is f'' positive, negative or zero?
(A) positive (B) negative (C) zero (D) cannot tell
Answer: A $f''(x) = 2x - 4$. At $x = 4$ it equals 4, which is positive.

Level 3. The table gives two values of f'. Estimate $f''(4.5)$ with the slope between them.
Answer: -5 Rise over run: $\dfrac{-34 - (-9)}{7 - 2} = \dfrac{-25}{5} = -5$.

Level 4. The table gives two values of f'. Estimate $f''(2.5)$ with the slope between them.
Answer: 6 Rise over run: $\dfrac{36 - (6)}{5 - 0} = \dfrac{30}{5} = 6$.

Second derivative of a curve

Level 1. The curve $x^2 + y^2 = 16$ passes through $(0, 4)$. Find $\dfrac{d^2y}{dx^2}$ at that point.
Answer: -0.25 $y' = -\dfrac{x}{y} = 0$ at the point. Then $y'' = -\dfrac{y - xy'}{y^2} = -\dfrac{y}{y^2} = -\dfrac{1}{y}$ there, which is -0.25.

Level 2. The curve $xy = 4$ passes through $(1, 4)$. Find $\dfrac{d^2y}{dx^2}$ at that point.
Answer: 8 $y' = -\dfrac{y}{x} = -4$ at the point. Differentiating again: $y'' = -\dfrac{y'x - y}{x^2} = 8$.

Level 3. The curve $x^3 + y^3 = 9$ passes through $(2, 1)$. Find $\dfrac{d^2y}{dx^2}$ at that point.
Answer: -36 Find y' first, differentiate it again (each y brings a new y'), then put in y' and the point. The answer is -36.

Level 4. The curve $x^3 + y^3 = 9$ passes through $(2, 1)$. Find $\dfrac{d^2y}{dx^2}$ at that point.
Answer: -36 Find y' first, differentiate it again (each y brings a new y'), then put in y' and the point. The answer is -36.

Common misconceptions

Students say	Say instead
f? is the same as $(f?)^2$.	f? is the derivative of f?, not its square.
The second derivative of e^(3x) is 3e^(3x).	Each derivative multiplies by 3: 9e^(3x).
sin x repeats after 2 derivatives.	sin, cos, –sin, –cos: it repeats after 4.
f? $>$ 0 means f is positive.	It means the slope of f is increasing. f can be negative.
In y? for an implicit curve, forgetting the new dy/dx from each y.	Differentiate dy/dx with the chain rule, then substitute dy/dx.